

— DOCTORAL RESEARCH PROPOSAL

Geometry- and Structure- Aware Graph Learning for *Large- Scale* Spatio-Temporal Interaction Networks

Integrating spatial geometry, topological structure, temporal dynamics, and Digital Twin principles into scalable graph neural network architectures for real-world infrastructure and interaction systems.

RESEARCH DOMAIN

Machine Learning · Graph Neural Networks

PROPOSED DURATION

3.5 – 4 Years (Full-time PhD)

APPLICATION AREAS

Infrastructure · Mobility · Digital Twins

DOCUMENT VERSION

Expanded — June 2026

Submitted for consideration · PhD Programme
Application

GNNs

Spatio-
Temporal

Digital
Twins

Graph
Learning

ABSTRACT

Research *Overview*

A concise summary of the research problem, proposed approach, and expected contributions.

Many real-world systems — including road networks, energy grids, water distribution systems, communication infrastructures, and mobility platforms — can be represented as **large-scale spatio-temporal interaction graphs**: networks where nodes occupy physical locations in space, edges encode relationships that change over time, and the geometry of the environment fundamentally shapes how influence, failure, and information propagate.

Despite major advances in Graph Neural Networks (GNNs), current architectures remain **geometry-agnostic and structure-agnostic**. They treat nodes as abstract symbols, ignore physical distances between nodes, discard directional relationships, and model time through crude discrete snapshots. This is a fundamental mismatch with real-world networks, where spatial distance governs interaction strength, directionality constrains flow, and topology evolves continuously.

This research proposes a unified theoretical and computational framework for **geometry- and structure-aware graph learning** that embeds spatial metrics, topological constraints, and temporal dynamics directly into graph representation and message-passing operators. Furthermore, this work introduces **Digital Twins** as a key application and validation context: the geometry-aware GNN framework developed here will serve as the computational intelligence layer inside Digital Twin systems for physical infrastructure networks, enabling real-time monitoring, predictive maintenance, and failure propagation modelling at scale.

The expected contributions include: (i) novel geometry-aware and direction-sensitive message-passing operators; (ii) continuous-time temporal graph extensions; (iii) theoretical expressiveness analysis; (iv) scalable implementation strategies; and (v) a Digital Twin integration architecture for infrastructure graph systems.



CORE RESEARCH QUESTION

How can spatial geometry, network topology, and temporal dynamics be embedded directly and systematically into graph learning operators to produce more expressive, accurate, and scalable models for real-world spatio-temporal interaction networks — and how can such models power next-generation Digital Twin systems for physical infrastructure?

Introduction & *Background*

The case for geometry-aware graph learning — what the field currently does, what it ignores, and why this matters for both science and real-world systems.

The Graph Learning Revolution

Graph Neural Networks have transformed machine learning across domains — from drug discovery and social network analysis to physics simulations and recommendation systems. The core idea is elegant: represent data as a graph of nodes and edges, then learn representations by iteratively aggregating information from each node's neighbours. This *message-passing* paradigm, formalised by Gilmer et al. (2017) and analysed theoretically by Xu et al. (2019), has proven remarkably effective.

Yet as GNNs are applied to increasingly real-world systems — road networks, power grids, sensor arrays, urban mobility infrastructure — a critical limitation emerges: **the geometry of the physical world is largely invisible to these models.**

The Geometry Blindness Problem

Consider a road network. Two intersections 500 metres apart behave fundamentally differently from two intersections 50 kilometres apart — even if they have the same number of connections. Traffic flows differently, failure propagates differently, congestion patterns differ. A standard GNN treats both cases identically: it sees only the graph topology, not the physical distance.

Similarly, in an energy grid, the direction of power flow along a transmission line matters enormously. In a water distribution network, elevation differences govern pressure and flow. In a mobility system, travel direction determines interaction patterns. Standard GNN message-passing is *permutation-invariant* — it aggregates neighbour features without regard to their spatial position or directional relationship.



What GNNs DO model well

Graph topology and connectivity; node attribute propagation; abstract relational reasoning; pattern matching in combinatorial structures.



What GNNs MISS in real systems

Physical distance between nodes; directional constraints on edges; spatial locality and anisotropy; continuous temporal evolution; geometric embedding of the network.

The Temporal Evolution Gap

Real networks are not static. Road networks experience time-varying congestion. Power grids adapt load dynamically. Sensor networks capture continuously evolving physical phenomena. Most GNN approaches handle temporal data by taking snapshots at discrete time steps — losing the continuous dynamics that drive real system behaviour. Continuous-time graph models remain an active and largely unsolved research frontier.

Why This Research, Why Now

Three convergent trends make this the right moment for this research:

1. **Data availability:** IoT sensor networks, satellite imagery, and connected infrastructure now generate rich, geolocated graph data at unprecedented scale.
2. **Digital Twin adoption:** Industry and government are investing heavily in Digital Twins — virtual replicas of physical systems — which require exactly the kind of geometry-aware, temporally evolving graph models this research proposes.
3. **Theoretical gaps:** Despite rapid empirical progress, the theoretical foundations of geometry-aware and temporally continuous graph learning remain under-explored — creating clear space for foundational PhD-level contributions.

Literature *Review*

A structured review of the key bodies of work this research builds upon, identifies gaps in, and aims to advance.

2.1 Foundational Graph Neural Networks

The modern GNN era begins with Kipf & Welling's (2017) Graph Convolutional Network (GCN), which showed that spectral convolution on graphs could be approximated by a simple neighbourhood aggregation rule. Veličković et al. (2018) introduced Graph Attention Networks (GAT), adding learnable attention weights to control each neighbour's contribution. Hamilton et al. (2017) proposed GraphSAGE, enabling inductive learning on unseen nodes through sampled neighbourhood aggregation.

Xu et al. (2019) provided the critical theoretical analysis, proving that the expressive power of message-passing GNNs is bounded by the Weisfeiler-Leman (WL) graph isomorphism test. This result reveals a fundamental ceiling on what standard GNNs can distinguish — motivating the search for architectures with greater structural expressiveness, which this research addresses through geometric enrichment.

2.2 Geometric and Equivariant Graph Learning

The field of geometric deep learning (Bronstein et al., 2021) provides the theoretical foundation for incorporating geometric structure into learning. Satorras et al. (2021) introduced $E(n)$ -Equivariant GNNs (EGNNs), ensuring that predictions are invariant or equivariant under rotations, translations, and reflections — essential for physical systems. Beaini et al. (2021) proposed Directional Graph Networks (DGN), using vector fields to introduce directional information into message passing.

Gasteiger et al. (2020) demonstrated that encoding directional information via angular embeddings significantly improves molecular property prediction — a result this research seeks to transfer to infrastructure and interaction network domains, where the geometry is Euclidean rather than molecular but the principle holds directly.

2.3 Temporal and Dynamic Graph Models

Modelling evolving graphs is an active research frontier. Xu et al. (2020) introduced the Temporal Graph Network (TGN) framework, using memory modules and temporal attention to handle continuous-time dynamic graphs — a key baseline for this research. Rossi et al. (2020) extended this to efficient streaming graph learning. Kazemi et al. (2020) provided a comprehensive survey of representation learning for dynamic graphs, identifying open problems including the lack of geometry-aware continuous-time models — exactly the gap this research addresses.

2.4 Spatio-Temporal Graph Models for Infrastructure

Traffic and infrastructure forecasting has produced a rich body of spatio-temporal GNN work. Li et al. (2018) pioneered Diffusion Convolutional Recurrent Neural Networks (DCRNN), modelling traffic as a diffusion process on road graphs. Wu et al. (2019) proposed Graph WaveNet, using adaptive adjacency matrices to capture hidden spatial dependencies without relying solely on physical proximity — a key motivation for learnable geometry kernels in this research. Zheng et al. (2020) demonstrated multi-attention mechanisms for urban traffic prediction at scale.

However, a common limitation across these works is their reliance on fixed, pre-defined graph structures that encode geography only implicitly. None simultaneously addresses geometry-awareness, direction-sensitivity, and continuous-time evolution within a unified theoretical framework.

2.5 Digital Twins and Graph-Based Intelligence

The Digital Twin concept — a live virtual replica of a physical system that evolves in synchrony with its real counterpart — was formalised by Grieves & Vickers (2017) and has since become a major industrial and research priority. Fuller et al. (2020) provide a comprehensive review of enabling technologies, identifying machine learning as a critical missing layer for intelligence inside Digital Twins. Rasheed et al. (2020) specifically highlight the need for models that capture spatial and temporal dynamics of physical systems — aligning precisely with the research proposed here.

Critically, while Digital Twin platforms exist for infrastructure (energy, water, transport), their AI components remain largely statistical or physics-based. **No existing work systematically integrates geometry-aware graph learning as the intelligence layer**

inside a Digital Twin for large-scale interaction networks — this represents a clear and significant research gap.



KEY RESEARCH GAP IDENTIFIED

No existing framework simultaneously integrates (i) geometry-aware message passing, (ii) direction-sensitive aggregation, (iii) continuous-time temporal evolution, and (iv) a scalable deployment architecture within Digital Twin systems. This proposal addresses all four gaps in a unified theoretical and applied framework.

CHAPTER 3

Problem *Statement*

The formal definition of what standard GNNs cannot do — and why this matters — explained both mathematically and in plain language.

Formal Definition

Let a spatio-temporal interaction network be defined as a time-varying attributed graph:

DEFINITION — SPATIO-TEMPORAL INTERACTION GRAPH

$$G_t = (V, E_t, X_t, S)$$

where:

- V = set of nodes (e.g. intersections, sensors, substations)
- $E_t \subseteq V \times V$ = time-dependent edge set (connections at time t)
- $X_t \in \mathbb{R}^{|V| \times d_x}$ = node feature matrix at time t
- $S : V \rightarrow \mathbb{R}^d$ = spatial embedding (each node has coordinates in d -dimensional space)

What Standard GNNs Compute

A standard message-passing GNN updates node embeddings at layer k as:

STANDARD GNN MESSAGE PASSING (GILMER ET AL., 2017)

$$h_i^{(k+1)} = \phi(h_i^{(k)}, \oplus_{j \in N(i)} \psi(h_i^{(k)}, h_j^{(k)}, e_{ij}))$$

where:

- $h_i^{(k)}$ = embedding of node i at layer k
- $N(i)$ = neighbours of node i
- ψ = message function (computes what each neighbour sends)
- $\oplus$ = permutation-invariant aggregation (sum, mean, max)
- ϕ = update function (updates node using aggregated messages)
- e_{ij} = edge features (if any)



PLAIN LANGUAGE EXPLANATION

Think of message passing like this: every node "listens" to all its neighbours and combines what it hears into an updated understanding of itself. The problem is that in a standard GNN, it doesn't matter whether a neighbour is 1 km away or 100 km away — or whether it is to the north or the south. The model treats all neighbours equally, regardless of where they are in space. For a social network, this might be fine. For a road network, power grid, or water system, it is a serious limitation.

Four Specific Failures

✗ Failure 1: Distance Blindness

The aggregation $\oplus$ treats a neighbour 500m away identically to one 500km away. Physical influence almost always decays with distance — this constraint is entirely absent.

✗ Failure 2: Direction Blindness

Permutation-invariant aggregation means it doesn't matter if a neighbour is "upstream" or "downstream", north or south. Anisotropic (direction-dependent) interactions are invisible.

✗ Failure 3: Static Structure

Most GNNs assume a fixed edge set E . Real networks evolve — roads close, grid connections change, sensors appear and disappear. Discrete snapshots miss continuous dynamics.

✗ Failure 4: Scale Collapse

Standard message passing scales quadratically with edges. Large infrastructure graphs (millions of nodes) overwhelm standard GNN implementations without specialised treatment.

Proposed Fix — Intuition

This research proposes augmenting the message function ψ with a **geometry-sensitive kernel** that modulates the influence of each neighbour based on: (a) how far away it is, (b) what direction it lies in, and (c) how recently or how long ago the interaction occurred. The result is a message-passing operator that is no longer blind to the physical world.

PROPOSED GEOMETRY-AWARE MESSAGE (CONCEPTUAL)

$$m_{ij}^{(k)} = K(d_{ij}, \theta_{ij}, \Delta t) \cdot \psi(h_i^{(k)}, h_j^{(k)})$$

where:

- d_{ij} = physical distance between node i and node j
- θ_{ij} = directional angle from i to j (orientation in space)
- Δt = time elapsed since last interaction
- $K(\cdot)$ = learnable geometry-sensitive kernel (the key novelty)



PLAIN LANGUAGE — WHAT K DOES

K is like a "weighting function" that asks: before node i listens to node j , how much should it trust what node j says? If j is very far away, K gives it less weight. If j is in the wrong direction (e.g., upstream in a water network when we care about downstream), K reduces its influence. If the interaction happened a long time ago, K also reduces it. This makes the model physically realistic in a way standard GNNs are not.

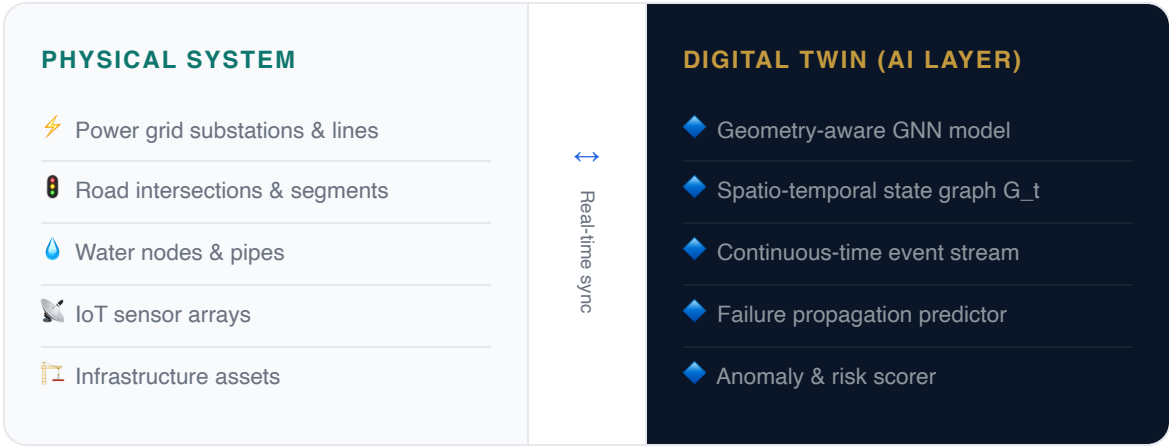
Digital Twins *Integration*

Why Digital Twins are a natural and powerful application context for this research — and how geometry-aware graph learning becomes the intelligence layer inside a Digital Twin.

What is a Digital Twin?

A Digital Twin is a **live, continuously updated virtual replica** of a physical system. It mirrors its real-world counterpart in real time, receiving sensor data, updating its internal state, generating predictions, and feeding insights back to operators. The concept, formalised by Grieves (2014), has grown into a major industrial paradigm — used in aviation (Boeing), manufacturing (Siemens), energy (National Grid), and smart cities.

A Digital Twin has three core layers: a *data layer* (sensor streams, IoT feeds), a *model layer* (the computational intelligence that interprets data), and an *interface layer* (visualisation and decision support). This research targets the **model layer** — arguing that geometry-aware spatio-temporal GNNs are the natural computational primitive for any Digital Twin whose physical counterpart is a networked system.



Why Geometry-Aware GNNs Are the Right Model for Digital Twins

Digital Twin intelligence systems must answer questions such as: "If substation X fails, which downstream areas lose power and in what order?" or "Given current traffic sensor readings,

where will congestion form in the next 20 minutes?" These questions are inherently spatial, directional, and temporal — precisely the dimensions that geometry-aware GNNs are designed to model.

Digital Twin Need	Geometry-Aware GNN Capability	Chapter
Spatial influence propagation	Distance-weighted kernel $K(d_{ij})$	Meth. Stage 1
Directional flow modelling	Orientation-sensitive kernel $K(\theta_{ij})$	Meth. Stage 2
Real-time state update	Continuous-time temporal extension	Meth. Stage 3
Failure cascade prediction	Structure-constrained message passing	Meth. Stage 2
City/grid scale operation	Spatial partitioning + sparsification	Meth. Stage 4

Why Digital Twins Strengthen the Proposal



Concrete Validation Context

Digital Twins provide real-world benchmarks that go beyond standard graph ML datasets. Testing geometry-aware GNNs inside a Digital Twin pipeline demonstrates practical value, not just academic improvement on benchmarks.



Funding & Industrial Relevance

UK Research and Innovation (UKRI), EPSRC, and Innovate UK fund Digital Twin research heavily. Framing this PhD within the Digital Twin paradigm significantly strengthens grant and industrial partnership prospects.



Novel Research Niche

No existing literature integrates geometry-aware graph learning as the computational intelligence layer inside Digital Twins. This positions the PhD at the frontier of two active fields simultaneously.



Broader Impact

Infrastructure Digital Twins have direct societal impact — smarter energy grids, safer road networks, more resilient water systems. This connects the fundamental ML research to national and global challenges.

**VERDICT: DIGITAL TWINS STRENGTHEN THIS PROPOSAL**

Integrating Digital Twins is recommended. It does not dilute the core ML research — it contextualises and validates it. The geometry-aware GNN framework proposed here is exactly the kind of model that infrastructure Digital Twins are missing. The integration adds a clear application narrative, strengthens industrial relevance, and opens additional funding avenues, while the core theoretical contributions remain fully intact.

CHAPTER 5

Research Objectives & *Questions*

Specific, measurable research objectives and the questions this PhD will answer.

Primary Research Aim



PRIMARY AIM

To develop, theoretically ground, and experimentally validate a unified framework of geometry- and structure-aware graph learning algorithms for large-scale spatio-temporal interaction networks, and to demonstrate their deployment as the intelligence layer within Digital Twin systems for physical infrastructure.

Specific Research Objectives

O1

Formal Geometric Representation

Theory

Develop a rigorous mathematical framework for representing spatio-temporal interaction networks as metric-embedded graphs, defining geometry-sensitive adjacency operators and characterising the geometric information that standard GNNs discard.

Research Question: What mathematical structures are required to fully represent geometric and topological constraints within a graph learning framework?

O2 Geometry-Aware Message-Passing Operators

Algorithmic

Design and implement novel message-passing operators that incorporate spatial distance, directional orientation, and structural constraints directly into the aggregation and update functions.

Research Question: What is the most expressive and computationally efficient form of geometry-aware message passing for spatially embedded graphs?

O3 Continuous-Time Temporal Graph Extension

Algorithmic

Extend the geometry-aware framework to handle continuously evolving graph topology and node states, moving beyond discrete snapshot approaches to continuous-time temporal models.

Research Question: How can spatial geometry and continuous-time dynamics be jointly modelled within a single, scalable graph learning framework?

O4 Theoretical Expressiveness Analysis

Theory

Formally analyse the expressive power of the proposed geometry-aware operators relative to standard GNNs and the Weisfeiler-Leman hierarchy; prove where and why geometry increases expressiveness.

Research Question: Does incorporating geometry strictly increase the class of graph properties that can be distinguished, and under what conditions?

Design efficient algorithms for large-scale deployment — including spatial indexing, graph sparsification, and distributed training — and demonstrate integration within a Digital Twin architecture for a real infrastructure network.

Research Question: How can geometry-aware graph learning be implemented at city or grid scale within a real-time Digital Twin pipeline?

Research *Methodology*

A four-stage methodology — from geometric graph representation to Digital Twin deployment — explained both formally and in plain language at each step.

1 Geometry-Aware Graph Representation

Months 1–10

The Core Idea

We want to move from treating the graph as an abstract network of dots and lines, to treating it as a network *embedded in physical space*. The first step is to build this spatial information into the edges of the graph.

Each edge (i, j) is augmented with a **geometry-sensitive kernel**:

GEOMETRY-SENSITIVE EDGE KERNEL

$$w_{ij} = K(d_{ij}, \theta_{ij}) = f_{\text{dist}}(d_{ij}) \cdot f_{\text{dir}}(\theta_{ij})$$

where:

- $d_{ij} = \|x_i - x_j\|$ = Euclidean (or geodesic) distance between nodes i and j
- $\theta_{ij} = \arctan2(y_j - y_i, x_j - x_i)$ = bearing angle from i to j
- $f_{\text{dist}}(\cdot)$ = learnable distance decay function (e.g., Gaussian, power law)
- $f_{\text{dir}}(\cdot)$ = learnable directional encoding (e.g., Fourier angular features)



PLAIN LANGUAGE — STAGE 1

Imagine you want to describe how strongly two towns influence each other. You would naturally say: nearby towns have more influence, and the direction matters (e.g., towns upstream in a river system affect downstream ones, not vice versa). Stage 1 builds exactly this intuition into the graph — turning each connection into a weighted, directional link whose strength reflects physical reality. The model learns the exact shape of this distance-decay and direction-bias from data.

This transforms the standard adjacency matrix $\mathbf{A}$ into a **metric-aware operator** $\hat{\mathbf{A}}$, where $\hat{A}_{ij} = A_{ij} \cdot K(d_{ij}, \theta_{ij})$. The graph structure now encodes geometry.

2 Structure-Constrained Message Passing

Months 6–18

The Core Idea

Using the geometry kernel from Stage 1, we redesign the message-passing equations so that the information each node receives from its neighbours is shaped by spatial distance, direction, and structural constraints.

GEOMETRY-AWARE MESSAGE-PASSING (FULL FORM)

$$m_{ij}^{(k)} = K(d_{ij}, \theta_{ij}) \cdot \psi(h_i^{(k)}, h_j^{(k)}, e_{ij})$$

$$h_i^{(k+1)} = \phi(h_i^{(k)}, \sum_{j \in N(i)} m_{ij}^{(k)})$$

Key properties of this formulation:

1. Spatial locality preserved: distant nodes contribute less by design
2. Anisotropy encoded: messages from different directions carry different weights
3. Topology-constrained: structural path constraints can be enforced via K
4. Differentiable: K is learnable end-to-end via gradient descent



PLAIN LANGUAGE — STAGE 2

Think of it like a telephone network. In a standard GNN, every person speaks to all their neighbours and each voice is equally loud. In our model, the volume of each voice is adjusted: neighbours who are far away speak more quietly (distance decay), neighbours who are "behind you" in the network flow speak more quietly than those "ahead of you" (directional weighting). The model learns exactly how loud each voice should be. The result is a much more realistic picture of how information, influence, or failure actually spreads through a physical network.

Methodology *Stages 3 & 4*

3 Continuous-Time Temporal Extension

Months 14–28

The Core Idea

Real networks evolve continuously — new edges appear and disappear, node states change every second. Instead of treating time as discrete steps (snapshot at $t=1$, $t=2$, $t=3...$), we model it as a continuous flow. The geometry kernel K is extended with a time dimension:

SPATIO-TEMPORAL KERNEL (CONTINUOUS TIME)

$$K(d_{ij}, \theta_{ij}, \Delta t) = f_{\text{dist}}(d_{ij}) \cdot f_{\text{dir}}(\theta_{ij}) \cdot f_{\text{time}}(\Delta t)$$

where $\Delta t = t_{\text{now}} - t_{\text{last_interaction}}(i, j)$

A natural parametric form (learnable):

$$K(d_{ij}, \Delta t) = \exp(-\alpha \cdot d_{ij}) \cdot \exp(-\beta \cdot \Delta t)$$

- α = spatial decay rate (how fast influence drops with distance)
- β = temporal decay rate (how fast past interactions become less relevant)
- Both α and β are learnable parameters, not hand-set constants



PLAIN LANGUAGE — STAGE 3

Imagine you are keeping track of road congestion. An accident that happened 30 minutes ago matters less than one that just happened — its effects have partially cleared. Stage 3 builds this "forgetting over time" directly into the model. The further in the past an interaction occurred, and the further away it happened in space, the less influence it has on the current prediction. The model learns how fast to forget — some systems have long memory (power grid faults), others have short memory (traffic jams). Instead of fixing these time windows manually, our model learns them from data.

This temporal extension draws on and extends the Temporal Graph Network (TGN) architecture of Rossi et al. (2020), incorporating spatial geometry as a first-class citizen in the temporal attention and memory update mechanisms.

4

Scalability, Systems Design & Digital Twin Integration

Months 24–42

The Scalability Challenge

A real road network has millions of intersections. A national power grid has hundreds of thousands of nodes. Naively applying geometry-aware message passing to graphs this large would be computationally prohibitive. Stage 4 addresses scalability through three strategies:

- **Spatial partitioning:** Divide the graph into geographically coherent subgraphs (using k-d trees or spatial indices). Process subgraphs in parallel, merging boundary information efficiently.
- **Geometry-guided sparsification:** Remove edges between nodes that are so far apart that $K(d_{ij}, \theta_{ij}) \approx 0$, reducing the effective graph density without information loss.
- **Distributed mini-batch training:** Adapt graph sampling methods (GraphSAINT, Cluster-GCN) to respect spatial locality when forming mini-batches, preserving geometric context.

Digital Twin Integration Architecture

The final stage deploys the trained geometry-aware GNN within a Digital Twin pipeline: sensor streams are ingested as continuous event sequences, mapped onto the spatio-temporal graph G_t , processed by the geometry-aware GNN, and outputs used for real-time state monitoring, anomaly detection, and predictive maintenance alerts.



PLAIN LANGUAGE — STAGE 4

Stages 1–3 build a very smart but potentially slow model. Stage 4 makes it fast enough to run on a real city-scale network in real time. Think of it like this: instead of trying to monitor the entire city at once, we divide it into neighbourhoods, monitor each neighbourhood efficiently, then share just the essential information between adjacent neighbourhoods. The result is a model that is both accurate (because it knows the geometry) and fast (because it exploits that geometry to cut unnecessary computation).

Evaluation Strategy

Task	Dataset	Metric	Baseline
Spatio-temporal forecasting	METR-LA, PEMS-BAY (traffic)	MAE, RMSE, MAPE	DCRNN, Graph WaveNet
Node classification	OGB-Proteins, spatial sensor graphs	ROC-AUC, F1	GCN, GAT, GraphSAGE
Link prediction (dynamic)	GDELT, ICEWS (event graphs)	MRR, Hits@K	TGN, JODIE
Anomaly detection	SWaT, WADI (industrial sensors)	Precision, Recall, F1	MTAD-GAT, GDN
Digital Twin validation	OpenStreetMap + simulated IoT	State estimation error	Physics-based DT baseline

Contributions & *Research Timeline*

The novel contributions this PhD aims to make, and a realistic four-year research plan.

Expected Original Contributions



Contribution 1 — Geometric Graph Theory

A rigorous mathematical framework for metric-embedded spatio-temporal graphs, including geometry-sensitive adjacency operators and their algebraic properties. This is a foundational theoretical contribution.



Contribution 2 — Novel GNN Operators

New distance- and direction-sensitive message-passing functions with learnable geometric kernels, demonstrated to outperform standard GNNs on spatio-temporal benchmarks.



Contribution 3 — Expressiveness Theory

Formal proofs showing that geometric enrichment strictly increases GNN expressiveness beyond the WL bound for geometrically embedded graphs — a significant theoretical result.



Contribution 4 — Continuous-Time Framework

Spatio-temporal kernel extending TGN-style continuous-time models with joint spatial-geometric awareness — a novel synthesis not achieved in existing literature.



Contribution 5 — Scalable Algorithms

Geometry-guided graph sparsification and spatial mini-batching strategies enabling city-scale and grid-scale deployment with theoretical complexity bounds.



Contribution 6 — Digital Twin Architecture

An end-to-end architecture integrating geometry-aware GNNs as the AI intelligence layer within a Digital Twin for infrastructure networks — demonstrated on a real case study.

Research Timeline



PHASE 1 — MONTHS 1–10

Literature Survey & Theoretical Framework

Comprehensive review of geometry-aware GNNs, temporal graph models, and Digital Twin literature. Formalise the spatio-temporal graph definition and geometry kernel theory. First conference paper: geometric graph representation framework.



PHASE 2 — MONTHS 6–18

Geometry-Aware Operator Design & Experiments

Implement and evaluate geometry-aware message-passing operators. Run experiments on traffic forecasting and node classification benchmarks. Second conference paper: geometry-aware GNN performance results.



PHASE 3 — MONTHS 14–28

Temporal Extension & Theoretical Analysis

Develop continuous-time spatio-temporal kernel. Complete expressiveness analysis and complexity proofs. Journal paper: unified spatio-temporal geometry-aware framework with theory.



PHASE 4 — MONTHS 24–38

Scalability & Digital Twin Integration

Develop spatial partitioning and sparsification strategies. Build and evaluate Digital Twin prototype on infrastructure dataset. Conference/journal paper: Digital Twin deployment results.



PHASE 5 — MONTHS 36–42

Thesis Writing & Submission

Consolidate all contributions into a coherent thesis. Internal review, corrections, and submission. Target: 4 publications (2 conference, 2 journal) by submission.

CHAPTER 8

Suggested Papers *to Review*

An annotated reading list in priority order — what each paper contributes and why it matters for this research. Start with Group A before moving to Groups B, C, and D.

Group A — Must Read First (GNN Foundations)

A1

How Powerful are Graph Neural Networks?

Xu, K., Hu, W., Leskovec, J., & Jegelka, S. · ICLR 2019

This is THE theoretical foundation paper for this research. It proves that standard GNNs are at most as powerful as the Weisfeiler-Leman graph isomorphism test — setting the theoretical ceiling this work aims to raise by incorporating geometry.

Core Theory

Read First

A2

Semi-Supervised Classification with Graph Convolutional Networks

Kipf, T.N. & Welling, M. · ICLR 2017

The paper that started the modern GNN era. Introduces the spectral convolution approximation that becomes the baseline all geometry-aware work is compared against. Understanding this is essential before reading anything else.

Foundational

Essential

A3

Graph Attention Networks

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. · ICLR 2018

Introduces learnable attention weights for neighbourhood aggregation — the direct precursor to the learnable geometric kernel this research proposes. Shows that not all neighbours should be treated equally, but only explores feature-space attention, not spatial attention.

Foundational

Baseline Model

A4

Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges

Bronstein, M.M., Bruna, J., Cohen, T., & Veličković, P. · arXiv 2021

The comprehensive theoretical manifesto for geometric deep learning — the field this research sits within. Provides the symmetry group formalism for understanding what geometric invariances and equivariances mean, and why they matter for physical systems.

Theory

Comprehensive

Group B — Geometry-Aware & Equivariant GNNs

B1

E(n) Equivariant Graph Neural Networks

Satorras, V.G., Hoogeboom, E., & Welling, M. · ICML 2021

Demonstrates how to make GNNs invariant/equivariant to rotations, translations, and reflections by incorporating relative positions and distances. This is the closest existing work to the geometry kernel proposed here — understand it deeply to differentiate your contribution.

Geometry-Aware

Closest Existing Work

B2

Directional Graph Networks

Beaini, D., Azizian, W., Wolf, G., & Hamilton, W. · ICML 2021

Introduces direction-sensitivity into GNNs using eigenvector-based flow fields. This directly addresses the "direction blindness" problem identified in the problem statement. Read alongside B1 to understand the landscape of directional approaches.

Directional

Key Related Work

B3

Directional Message Passing for Molecular Graphs

Gasteiger, J., Groß, J., & Günnemann, S. · ICLR 2020

Shows that encoding distance and angular (directional) information between atoms dramatically improves predictions on molecular graphs. The domain is chemistry, but the geometric principle — that direction matters as much as distance — applies directly to infrastructure networks.

Directional

Distance Kernels

Group C — Temporal & Dynamic Graph Models

C1

Temporal Graph Networks for Deep Learning on Dynamic Graphs

Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. · ICML Workshop 2020

The most comprehensive continuous-time dynamic graph learning framework — the key temporal baseline this research extends with spatial geometry. Understand the memory module and temporal attention mechanism thoroughly.

Temporal GNN

Key Baseline

C2

Inductive Representation Learning on Temporal Graphs

Xu, D., Ruan, C., Körpeoglu, E., Kumar, S., & Achan, K. · ICLR 2020

Introduces temporal attention for continuous-time graphs — a direct component this research integrates with geometric awareness. TGAT (Temporal Graph ATtention) is an important baseline for the temporal experiments.

Temporal GNN

Baseline

C3

Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting

Li, Y., Yu, R., Shahabi, C., & Liu, Y. · ICLR 2018

Pioneered spatio-temporal GNN for traffic — models traffic flow as a diffusion process on a road graph, encoding some spatial structure via the diffusion operator. The primary real-world application baseline for this research.

Spatio-Temporal

Traffic

Key Baseline

C4

Graph WaveNet for Deep Spatial-Temporal Graph Modeling

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Philip, S.Y. · IJCAI 2019

Introduces learnable adaptive adjacency matrices — the graph learns its own spatial dependencies from data rather than using predefined distance thresholds. A direct predecessor to the learnable geometry kernel proposed here.

Spatio-Temporal

Learnable Adjacency

C5

Representation Learning for Dynamic Graphs: A Survey

Kazemi, S.M., Goel, R., Jain, K., et al. · JMLR 2020

Comprehensive survey of all approaches to dynamic graph learning. Essential for understanding the taxonomy of methods, what has been tried, and what gaps remain. Maps directly to the temporal extension objective of this research.

Survey

Essential Reading

Group D — Digital Twins & Infrastructure

D1

Digital Twin: Enabling Technologies, Challenges and Open Research

Fuller, A., Fan, Z., Day, C., & Barlow, C. · IEEE Access, 2020

The most-cited comprehensive review of Digital Twin technologies. Critically, identifies machine learning as a missing intelligence layer in current DT implementations — this is the exact gap this research fills.

Digital Twin

Key Framing Paper

D2

Digital Twin: Values, Challenges and Enablers from a Modeling Perspective

Rasheed, A., San, O., & Kvamsdal, T. · IEEE Access, 2020

Analyses what mathematical modelling approaches are needed inside Digital Twins for physical systems — specifically highlights the need for spatio-temporal models that go beyond current physics-based approaches. Directly motivates the GNN approach proposed here.

Digital Twin

Modelling

D3

Graph Neural Network-Based Anomaly Detection in Multivariate Time Series

Deng, A. & Hooi, B. · AAAI 2021

Demonstrates GNN-based anomaly detection on industrial sensor networks — exactly the kind of Digital Twin monitoring task (Stage 4) this research will tackle. The graph structure is learned rather than predefined, which this work improves by incorporating geometric priors.

Digital Twin

Anomaly Detection

Group E — Scalability

E1

Cluster-GCN: An Efficient Algorithm for Training Deep and Large Graph Convolutional Networks

Chiang, W., Liu, X., Si, S., Li, Y., Bengio, S., & Hsieh, C.J. · KDD 2019

Introduces graph clustering for efficient mini-batch training — the starting point for the spatial partitioning strategy in Stage 4. The geometry-guided version proposed in this research adapts their cluster definition to respect spatial locality.

Scalability

Mini-batching

E2

GraphSAINT: Graph Sampling Based Inductive Learning Method

Zeng, H., Zhou, H., Srivastava, A., Kannan, R., & Prasanna, V. · ICLR 2020

State-of-the-art graph sampling for large-scale GNN training, with theoretical guarantees on bias. The spatial mini-batching strategy in Stage 4 builds directly on this framework, adding geometric locality constraints to the sampler.

Scalability

Large-Scale

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